

ANDREY OSIPOV

Python Backend Engineer

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GENERAL INFORMATION

Target role: Python Developer

Specialization: Software developer

Employment: Full-time

Work format: Remote, hybrid, on-site

Business trips: Not available

Location: Belgrade, Serbia

Citizenship: Russia

Gender: Male

SUMMARY

Python backend engineer with 3+ years building and shipping production services. Specialized in high-load systems on FastAPI and Django – performance optimization, large-scale data (SQL and NoSQL), testing, observability, and CI/CD. Comfortable owning services end to end, from database schema and API design to infrastructure and deployment. Background in information security (MEPhI) – I write code with security and load in mind.

EXPERIENCE

Total commercial experience – 3 years 5 months.

Safe Transport Innovation Center

Sep 2025 – Present

Python Backend Engineer

Remote

High-load services for transport-security systems; multi-terabyte MongoDB (largest collection 7.2B records), message brokers, Redis, FastAPI, Docker; heavy parallel data processing; sensitive data under NDA.

- Designed and shipped an aggregating collection over a multi-terabyte MongoDB store, consolidated dozens of disparate collections into a single model, cut time to retrieve a full record summary from 3–5s to <100ms and reduced database load by ~40%.
- Built from scratch a Git-style state-versioning system (change commits, state diffs, stable versions and a history graph), made regressions between releases traceable and halved acceptance-testing time.
- Stabilized and extended a high-load parallel data-processing system (threading, multiprocessing) and ensured correct service behavior under peak load.
- Introduced centralized error monitoring (Sentry) and uptime alerting to Telegram across my services, cut mean time to detect (MTTD) from hours to minutes and eliminated silent failures.
- Rewrote legacy services on async FastAPI, reduced p95 latency from ~500ms to 80ms and increased throughput 1.5–2x.
- Applied information-security practices in services handling sensitive data under NDA: input validation, access control and dependency review.

Stack: Python, FastAPI, MongoDB, Redis, Docker, message brokers, Sentry

Rutube

Sep 2024 – Aug 2025

Python Backend Engineer

Moscow

Leading Russian video-hosting platform. Django (DRF) monolith plus aiohttp and FastAPI microservices; team of 15 engineers; owned my services end to end, from development and code review to running them in production.

- Optimized 10+ high-load API endpoints using Prometheus and Grafana data: reworked ORM queries (select_related/prefetch, indexes), added Redis caching and rate limiting and cut p95 latency from ~1s to <100ms at ~100 RPS per instance.
- Rewrote 10+ minute SQL queries using Django CTEs and window functions, cut their execution time to seconds and removed critical load from the database.
- Moved 200+ tests to pytest with parallelized CI runs, raised coverage from below 50% and sped up the pipeline.

- Migrated API docs to OpenAPI 3.0, replaced django-rest-swagger with drf-spectacular, documented 150+ endpoints and delivered clear API contracts to neighboring teams.
- Ran a security audit of the services, closed SQL injections and replaced insecure libraries.
- Unified Dockerfiles for two services, adopted the uv package manager and sped up image builds.

Stack: Python, Django, DRF, aiohttp, FastAPI, PostgreSQL, Redis, Prometheus, Grafana, Docker, pytest

PSK SOFT LLC

May 2023 – Aug 2024

Python Developer

Moscow

Backend development of corporate web services on Python/Django with PostgreSQL: server-side logic, integrations, and production infrastructure.

- Built and shipped the backend of 4 corporate web projects on Python/Django (DRF) with PostgreSQL: domain models, business logic, catalogs, and admin panels.
- Implemented server-side handling of client requests with validation, PostgreSQL storage and instant notifications and cut business response time from hours to 1–2 minutes.
- Integrated payment systems, web analytics and email delivery via APIs and ensured reliable payment handling.
- Optimized service performance: added caching, reworked SQL queries and indexes and sped up page response.
- Set up production infrastructure and deployment (hosting, domains, SSL/HTTPS) and kept services running reliably in production.

Stack: Python, Django, DRF, PostgreSQL, Nginx, Docker

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, Django, Django REST Framework, aiohttp, SQLAlchemy, Alembic, Pydantic, Celery, REST API, gRPC, OpenAPI/Swagger, JWT/OAuth

Databases: PostgreSQL, MongoDB, Redis, ClickHouse, Elasticsearch, Apache Cassandra

Messaging / Streaming: Apache Kafka, RabbitMQ

DevOps & Infra: Docker, Docker Compose, Kubernetes, Nginx, CI/CD (GitHub Actions, GitLab CI), Linux, S3

Monitoring: Prometheus, Grafana, Sentry

Practices: Pytest, Git, Agile, microservices, asyncio, concurrency (threading, multiprocessing)

EDUCATION

Higher education – BSc, Information Security – National Research Nuclear University MEPhI, Moscow 2025
Faculty of Cybernetics and Information Security

LANGUAGES

Russian – Native English – B2+ (Upper-Intermediate), certified

ADDITIONAL INFORMATION

References: Rutube – Mikhail, Head of Backend Development